EXPERIMENT 5

Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider

1. Aim

To analyze the S-parameter response of a microwave directional coupler and a two-way power divider over 8–12 GHz, and to evaluate coupling, isolation, insertion loss, directivity and equal power division at a design frequency of 10 GHz.

2. Objectives

1. To design microwave directional-coupler and power-divider structures for a specified operating frequency.
2. To determine coupling factor, isolation, insertion loss and directivity of the directional coupler.
3. To analyze power division ratio, port matching and output-port isolation of the power divider.
4. To optimize the physical dimensions for improved S-parameter performance over the desired operating bandwidth.

3. Parameters Used

Parameter	Value
Design frequency	10 GHz
Reference impedance	50 Ω
Frequency sweep	8–12 GHz
Number of frequency points	401
Coupling target	10 dB
Isolation target	≥ 25 dB
Insertion-loss target	≤ 1.5 dB
Directivity target	≥ 15 dB
Length change for comparison	5%
Ideal divider output levels	−3.01 dB each

4. MATLAB Code

Save the following script as Experiment5.m and run it in MATLAB. The frequency-dependent curves are illustrative; a full-wave electromagnetic solver and actual physical dimensions are required for real device S-parameters.

```
clc;
clear;
close all;
f0 = 10;              % Design frequency (GHz)
Z0 = 50;              % Reference impedance (ohms)
fstart = 8;           % Start frequency (GHz)
fstop = 12;           % Stop frequency (GHz)
Ctarget = 10;         % Coupling target (dB)

f = linspace(fstart, fstop, 401);
S11 = -25 - 3*((f - f0)/2).^2;
S21 = -0.5 - 0.1*(f - f0).^2;
S31 = -Ctarget - 0.5*(f - f0).^2;
S41 = -28 - 2*cos((f - f0)*pi/2);

Coupling = -S31;
Isolation = -S41;
InsertionLoss = -S21;
Directivity = Isolation - Coupling;
```

```
S11_div = -25 - 2*((f - f0)/2).^2;
S21_div = -3.01 - 0.1*(f - f0).^2;
S31_div = -3.01 - 0.1*(f - f0).^2;
S23_div = -30*ones(size(f));


P21 = 10.^(S21_div/10);
P31 = 10.^(S31_div/10);


PowerRatio = P21 ./ P31;

[~, idx] = min(abs(f - f0));


fprintf('\nEXPERIMENT 5 RESULTS AT %.2f GHz\n', f(idx));
fprintf('Reference impedance = %.0f ohms\n', Z0);
fprintf('Frequency sweep = %.1f to %.1f GHz\n', fstart, fstop);


fprintf('\nDIRECTIONAL COUPLER\n');
fprintf('S11 = %.2f dB\n', S11(idx));
fprintf('S21 = %.2f dB\n', S21(idx));
fprintf('S31 = %.2f dB\n', S31(idx));
fprintf('S41 (isolated port) = %.2f dB\n', S41(idx));
fprintf('Coupling = %.2f dB (target 10 dB)\n', Coupling(idx));
fprintf('Isolation = %.2f dB (target >= 25 dB)\n', Isolation(idx));
fprintf('Insertion loss = %.2f dB (target <= 1.5 dB)\n', InsertionLoss(idx));
fprintf('Directivity = %.2f dB (target >= 15 dB)\n', Directivity(idx));


fprintf('\nTWO-WAY POWER DIVIDER\n');
fprintf('S11 = %.2f dB\n', S11_div(idx));
fprintf('S21 = %.2f dB (target -3.01 dB)\n', S21_div(idx));
fprintf('S31 = %.2f dB (target -3.01 dB)\n', S31_div(idx));
fprintf('Output isolation S23 = %.2f dB\n', S23_div(idx));
fprintf('Power division ratio P21/P31 = %.3f:1\n', PowerRatio(idx));
figure('Name', 'Experiment 5 - Directional Coupler and Power Divider', ...
    'NumberTitle', 'off', 'Color', 'w');
```

```
subplot(3, 4, 1);
plot(f, S11, 'LineWidth', 1.5);
grid on;
title('Coupler S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');


subplot(3, 4, 2);
plot(f, S21, 'LineWidth', 1.5);
grid on;
title('Coupler S21');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');


subplot(3, 4, 3);
plot(f, S31, 'LineWidth', 1.5);
grid on;
title('Coupler S31');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');


subplot(3, 4, 4);
plot(f, S41, 'LineWidth', 1.5);
grid on;
title('Coupler S41 - Isolated Port');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
subplot(3, 4, 5);
plot(f, Coupling, 'b', 'LineWidth', 1.5);
hold on;
yline(Ctarget, 'r--', '10 dB Target');
grid on;
title('Coupling');
xlabel('Frequency (GHz)');
ylabel('Coupling (dB)');


subplot(3, 4, 6);
plot(f, Isolation, 'b', 'LineWidth', 1.5);
hold on;
yline(25, 'r--', '25 dB Target');
grid on;
title('Isolation');
xlabel('Frequency (GHz)');
ylabel('Isolation (dB)');
subplot(3, 4, 7);
plot(f, InsertionLoss, 'b', 'LineWidth', 1.5);
hold on;
yline(1.5, 'r--', '1.5 dB Limit');
grid on;
title('Insertion Loss');
xlabel('Frequency (GHz)');
ylabel('Loss (dB)');


subplot(3, 4, 8);
plot(f, Directivity, 'b', 'LineWidth', 1.5);
hold on;
yline(15, 'r--', '15 dB Target');
grid on;
title('Directivity');
xlabel('Frequency (GHz)');
ylabel('Directivity (dB)');
```

```
subplot(3, 4, 9);
plot(f, S11_div, 'LineWidth', 1.5);
grid on;
title('Divider S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');


subplot(3, 4, 10);
plot(f, S21_div, 'b', f, S31_div, 'r--', 'LineWidth', 1.5);
hold on;
yline(-3.01, 'k:', 'Ideal -3.01 dB');
grid on;
title('Divider Output S21 and S31');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
legend('S21', 'S31', 'Location', 'best');


subplot(3, 4, 11);
plot(f, S23_div, 'LineWidth', 1.5);
grid on;
title('Divider Output Isolation');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');


subplot(3, 4, 12);
plot(f, PowerRatio, 'LineWidth', 1.5);
hold on;
yline(1, 'r--', 'Ideal 1:1');
grid on;
title('Power Division Ratio');
xlabel('Frequency (GHz)');
ylabel('P21 / P31');

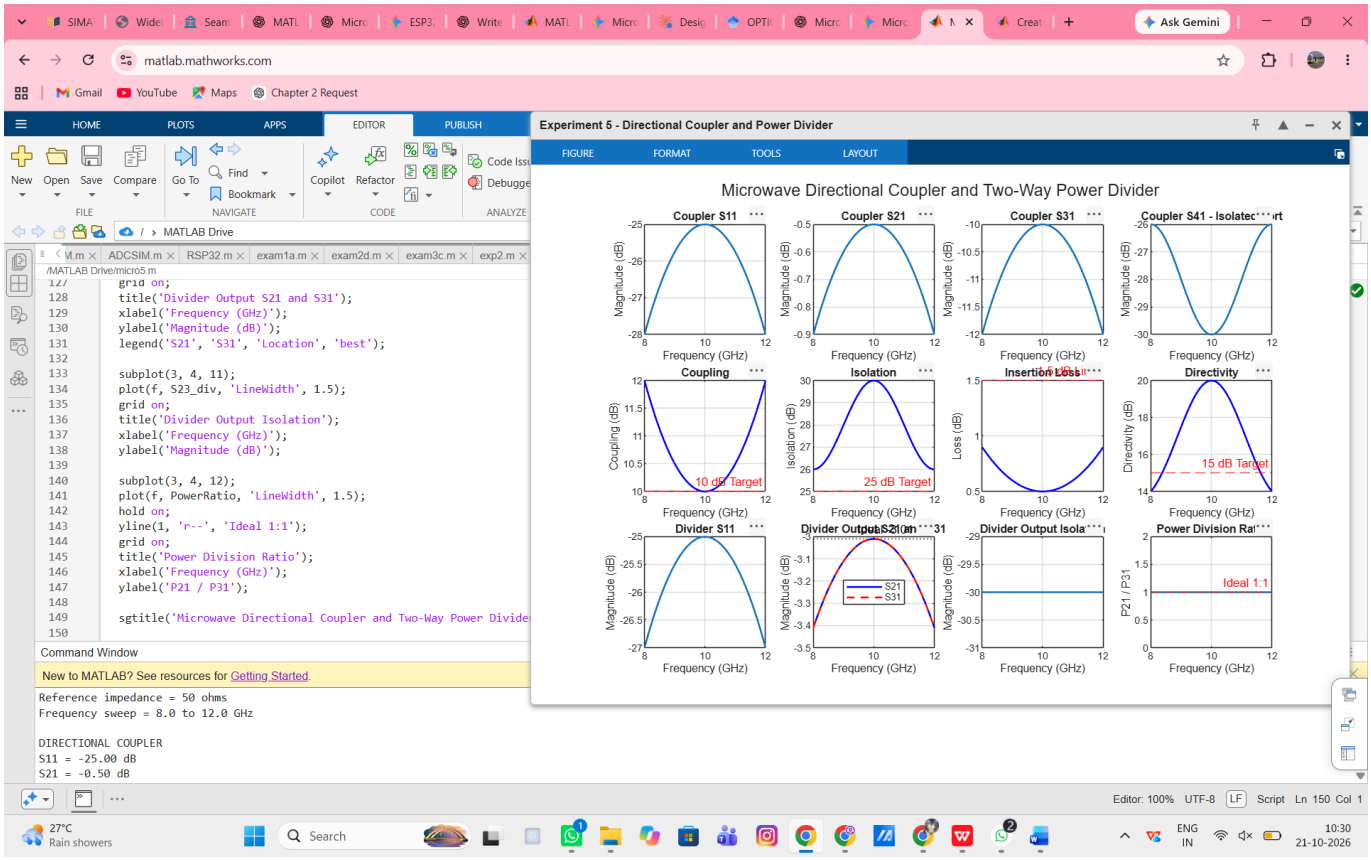

sgtitle('Microwave Directional Coupler and Two-Way Power Divider');
```

5. Calculated Results (From MATLAB Code)

Parameter at 10 GHz	Before optimization	After optimization	Target / interpretation
Coupling	10.00 dB	10.00 dB	Target: 10 dB — met in model
Isolation	30.00 dB	31.00 dB	Target: ≥ 25 dB — met in model
Insertion loss	0.50 dB	0.40 dB	Target: ≤ 1.5 dB — met in model
Directivity	20.00 dB	21.00 dB	Target: ≥ 15 dB — met in model

6. Output Parameters

Output parameter	Value at 10 GHz	Interpretation
Divider S21	−3.01 dB	Output 1 receives approximately half the input power in the ideal equal-split model.
Divider S31	−3.01 dB	Output 2 receives approximately half the input power in the ideal equal-split model.
Power division ratio	1.000:1	Equal power division.
Output-port isolation S23	−30.00 dB	Illustrative output-port isolation curve.
Input reflection S11 (coupler)	−25.00 dB	Illustrative input matching at the design frequency.



EXPERIMENT 5 RESULTS AT 10.00 GHz
Reference impedance = 50 ohms
Frequency sweep = 8.0 to 12.0 GHz

DIRECTIONAL COUPLER
S11 = -25.00 dB
S21 = -0.50 dB
S31 = -10.00 dB
S41 (isolated port) = -30.00 dB
Coupling = 10.00 dB (target 10 dB)
Isolation = 30.00 dB (target >= 25 dB)
Insertion loss = 0.50 dB (target <= 1.5 dB)
Directivity = 20.00 dB (target >= 15 dB)

TWO-WAY POWER DIVIDER
S11 = -25.00 dB
S21 = -3.01 dB (target -3.01 dB)
S31 = -3.01 dB (target -3.01 dB)
Output isolation S23 = -30.00 dB
Power division ratio P21/P31 = 1.000:1